CIS 635 Knowledge Discovery & Data Mining

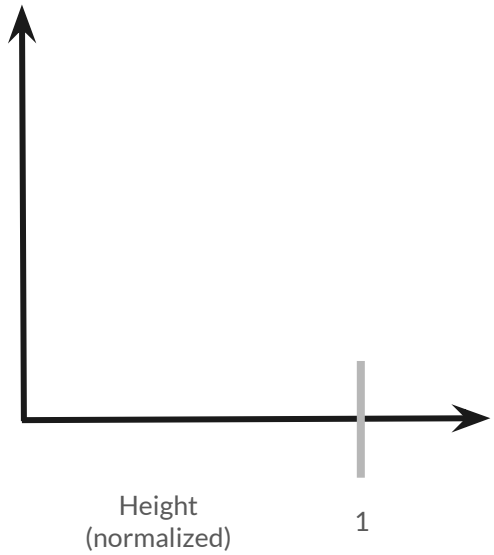
- Curse of Dimensionality
- Principal Component Analysis (PCA)



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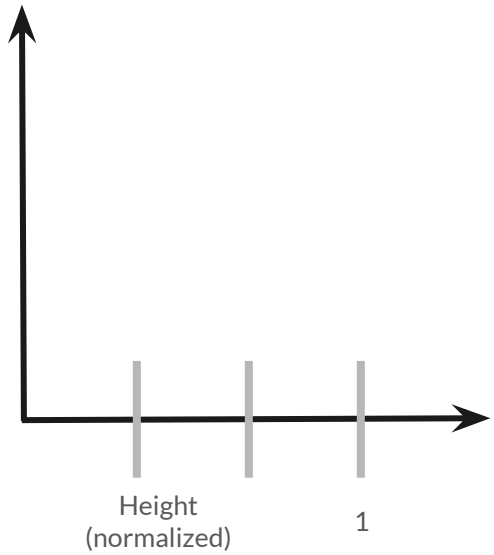
Curse of Dimensionality

- x axis : Heights (min-max normalized: [0-1]



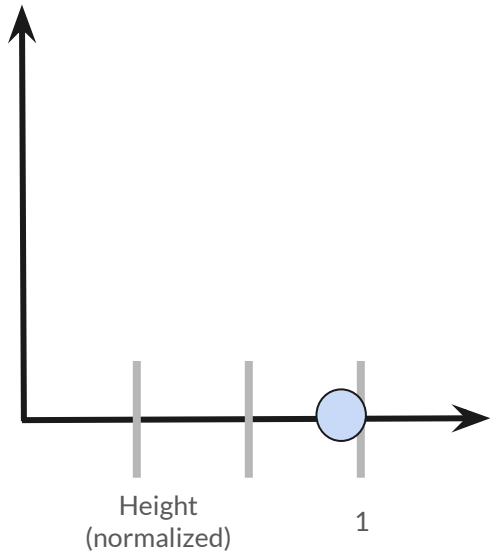
Curse of Dimensionality

- x axis : Heights (min-max normalized: [0-1]
- **3 equal bins (aka numeric to categorical)**



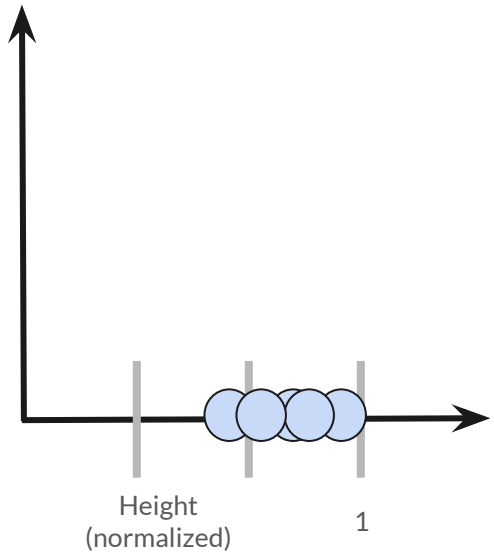
Curse of Dimensionality

- x axis : Heights (min-max normalized: [0-1]
- 3 equal bins (aka numeric to categorical)
- **The tallest person in the class**



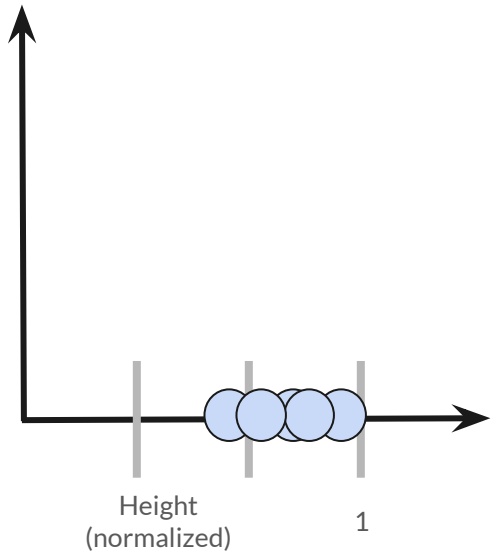
Curse of Dimensionality

- x axis : Heights (min-max normalized: [0-1]
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- **All of us**



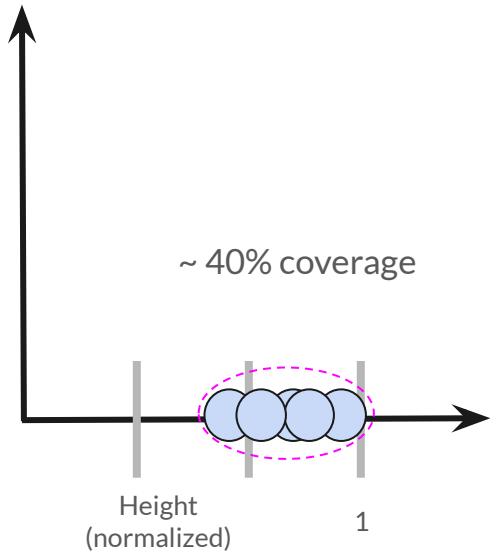
Curse of Dimensionality

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- **I don't think we have someone with 0.5 times height than the tallest**



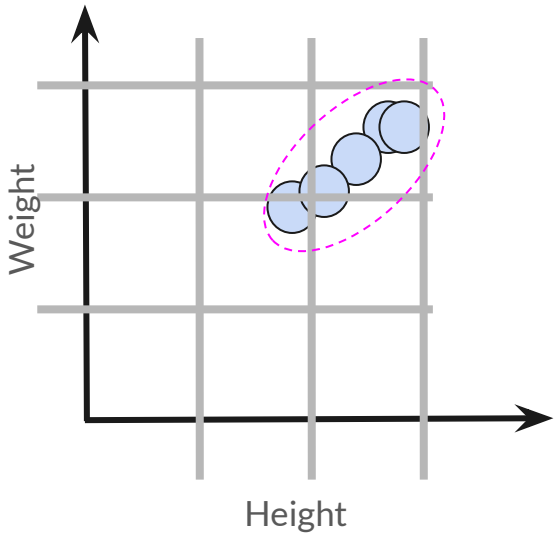
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- x axis : Heights (min-max normalized: [0-1]
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- I don't think we have someone with 0.5 times height than the tallest
- **Is ~40% domain coverage a reasonable assumption?**



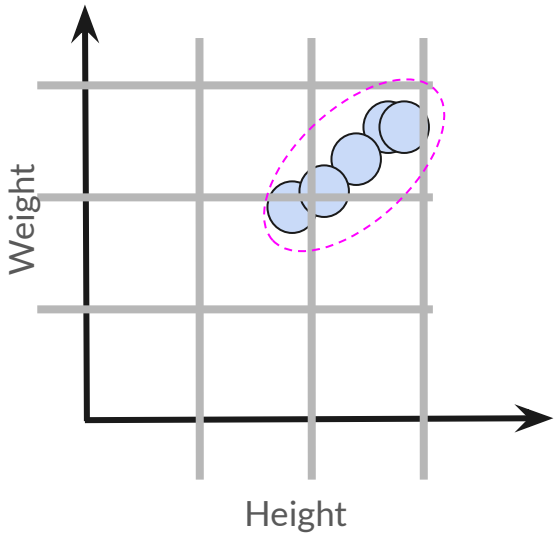
Curse of Dimensionality

- We have added Weight as our second feature dimension
- Do you find the samples represent our class?
- What's the domain coverage now?



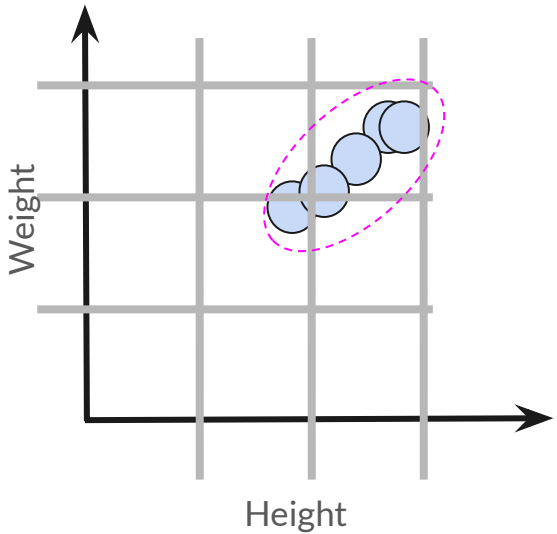
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- **Is ~10% a reasonable assumption?**



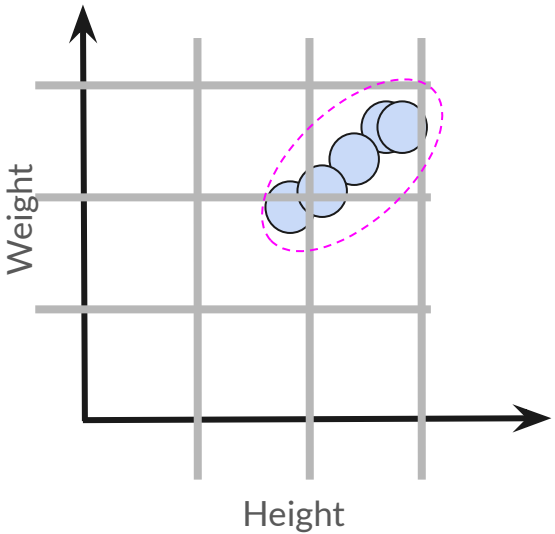
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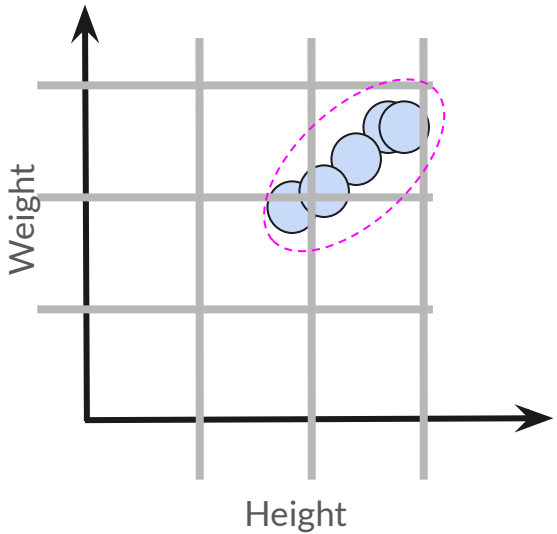
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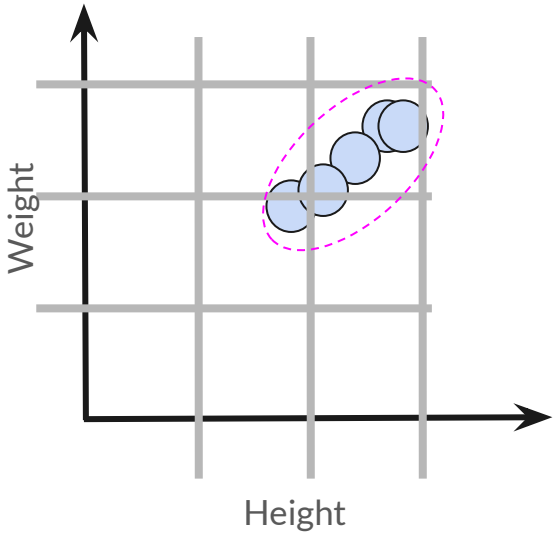
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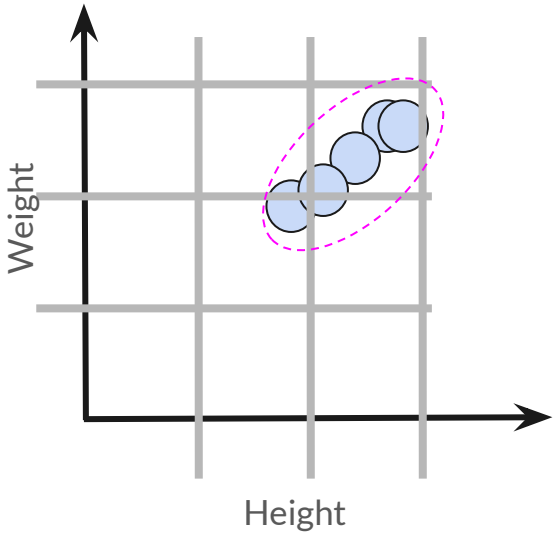
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- **What's the number would look like if we add 2 more dimensions?**

- Empty space grows exponentially with the increase in adding new features.
- **Data distribution becomes sparse, and difficult to learn a good model.**



Dimensionality Reduction

General idea

- You have some data of feature dimension size, $|D|$

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- You have some data of feature dimension size, $|D|$
- You want to compress them to of size, $|d|$
- $|d| < |D|$

Dimensionality Reduction

- Linear
  - Principal Component Analysis (PCA)
  - SVD
- Neural Networks
  - Auto Encoders
  - RBMs

Covariance

- Variance relationship between a pair of variables

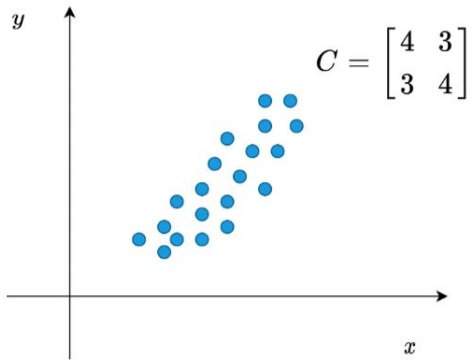
$$cov_{x,y} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{N - 1}$$

Covariance

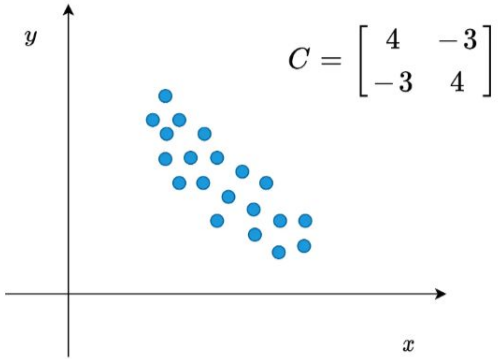
- Variance relationship between a pair of variables
- It's a symmetric matrix; right?

$$cov_{x,y} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{N - 1}$$

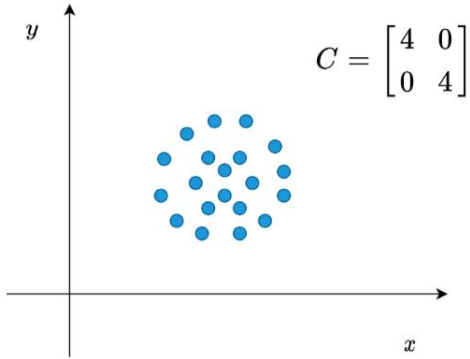
Positive Covariance



Negative Covariance

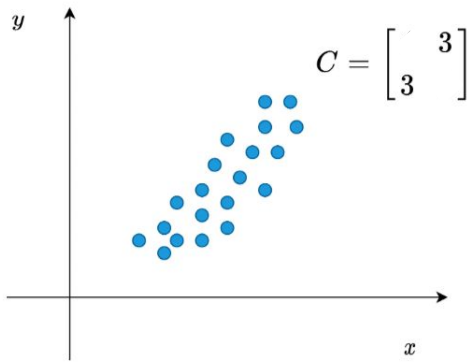


Zero Covariance

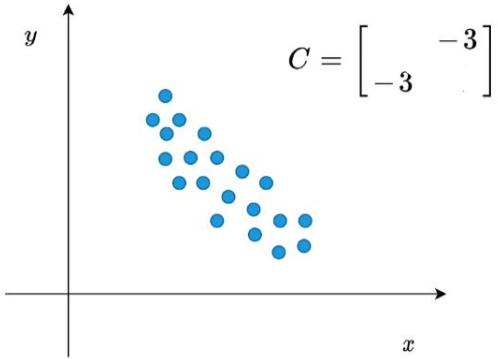


[5 Things you should know about Covariance]

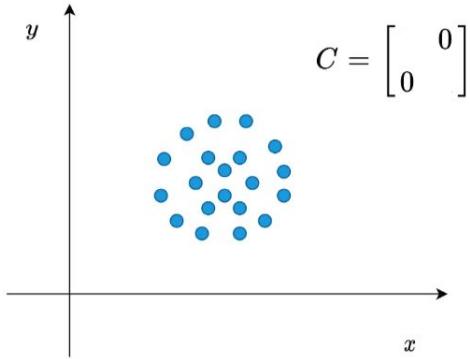
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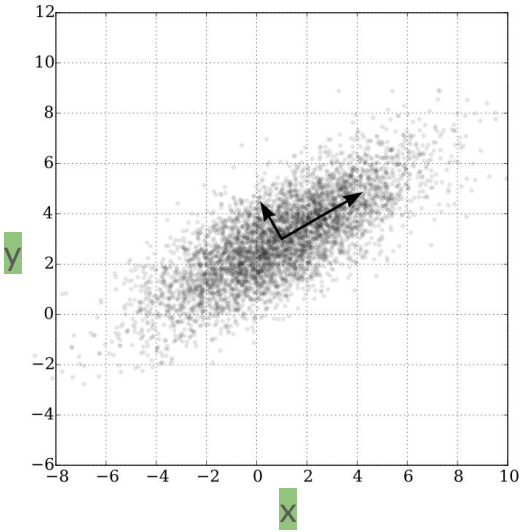
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Principal Component Analysis (PCA)

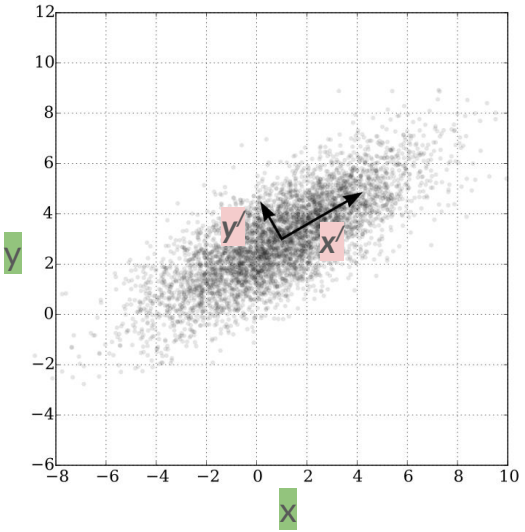
General idea: 2D Gaussian example

- Features x and y shows some relationships



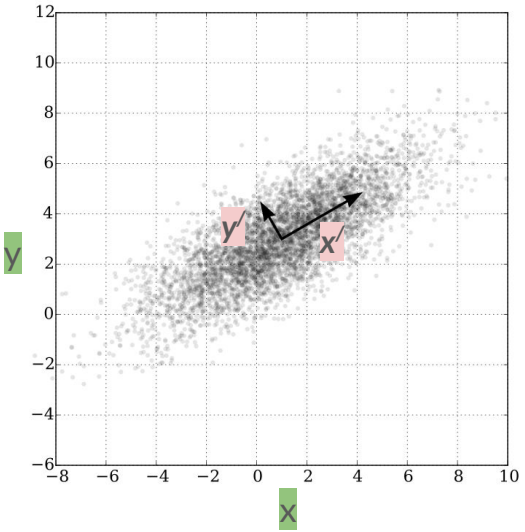
General idea: 2D Gaussian example

- Features x and y shows some relationships
- This 2D Gaussian has its own coordinates (off the reference cartesian coordinates x' and y'; right?)



General idea: 2D Gaussian example

- Features *x* and *y* shows some relationships
- This 2D Gaussian has its own coordinates (off the reference cartesian coordinates x' and y'; right?)
- **The principal components**



How to ..

PCA steps

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4. Compute Eigenvalues and Eigenvectors of the Covariance Matrix by solving the linear dynamical system at the right

$$AX = \lambda X$$

$$AX - \lambda X = 0$$

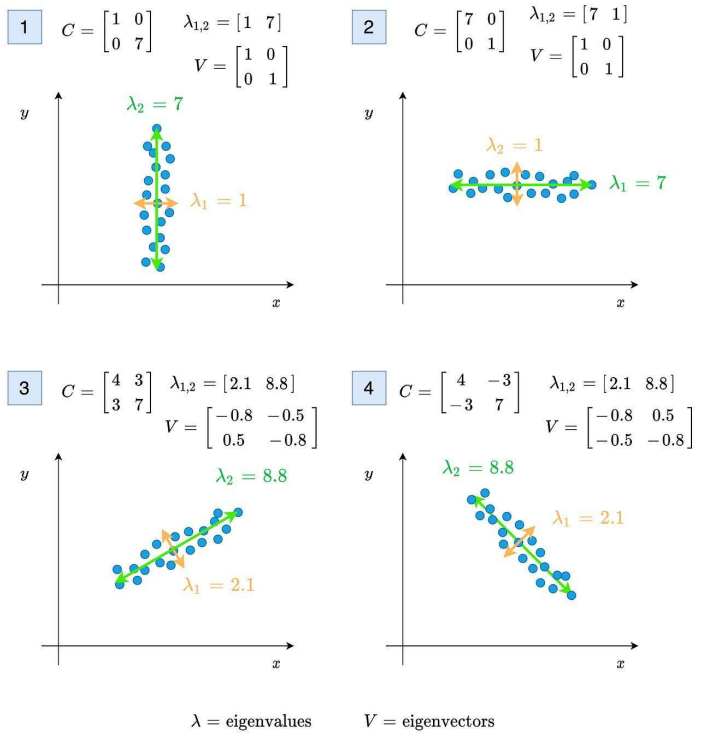
or

$$(A - \lambda I)\,X = 0$$

If ***A*** is a *nxn* matrix, solving this linear dynamical system will give *n* **eigenvalues**, and n associated *n* **eigenvectors**

PCA

[Notebook presentation]



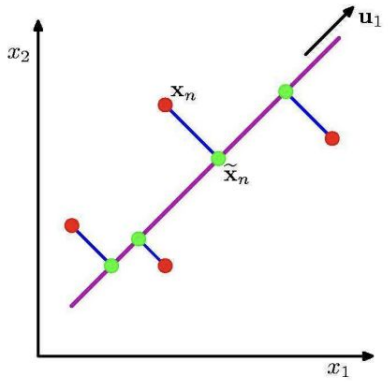
[ref]

What are we optimizing?

- 2D example (red points)
- Green points are 1D projections/transformations
- We are reducing data definitions from 2D to 1D; this can be generalized from D to M dimensions

Two techniques (in general):

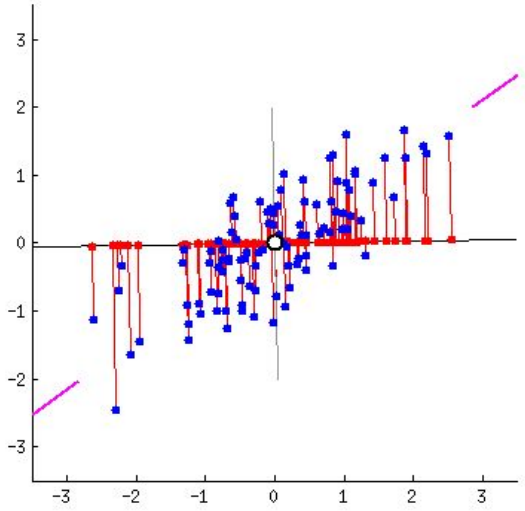
- Maximize variance (distance between each green-red paris)
- Minimize errors



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Two techniques (in general):

- Maximize variance
- Minimize errors (distance between each green-red paris)

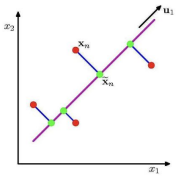


ref

What are we optimizing?

Standard PCA: Variance maximization

- One dimensional example
- Objective: maximize projected variance w.r.t. $\mathbf{U}_1$



$$\frac{1}{N}\sum_{n=1}^{N}\{\mathbf{u}_1^T\mathbf{x}_n - \mathbf{u}_1^T\bar{\mathbf{x}}\}^2 = \mathbf{u}_1^T\mathbf{S}\mathbf{u}_1$$

  - where sample mean and data covariance are:

$$\bar{\mathbf{x}} = \frac{1}{N}\sum_{n=1}^{N}\mathbf{x}_n$$

$$\mathbf{S} = \frac{1}{N}\sum_{n=1}^{N}(\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T$$

- Must constrain $\|\mathbf{u}_1\|$: via Lagrange multiplier, maximize w.r.t $\mathbf{u}_1$

$$\mathbf{u}_1^T\mathbf{S}\mathbf{u}_1 + \lambda(1 - \mathbf{u}_1^T\mathbf{u}_1)$$

- Optimal $\mathbf{u}_1$ is principal component (eigenvector with maximal eigenvalue)

ref